Chapter 8: Gravitation

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1 Worked Examples

1.1 Newton's Discovery of the Inverse-Square Law

In the 17th century, Johannes Kepler established that the orbital period T of a planet and the semi-major axis a of its elliptical orbit satisfy Kepler's third law,

$$T^2 \propto a^3. \tag{1}$$

According to his later account, Isaac Newton used Kepler's third law and the dynamics of circular motion to infer how the force maintaining a planet's orbit depends on its distance from the Sun.

We will reconstruct Newton's reasoning by approximating planetary orbits as circles. Consider a planet of mass m moving in a circular orbit of radius r around the Sun, completing one revolution in a period T.

(a) Express the orbital speed v of the planet in terms of r and T.

(b) Using Newton's second law, determine the magnitude of the force required to maintain the planet's circular motion.

(c) Using Kepler's third law, show that the force maintaining the planet's orbit is inversely proportional to the square of its distance from the Sun,

$$F(r) \propto \frac{1}{r^2}, \tag{2}$$

for a planet of fixed mass.

(d) Newton also investigated whether the force that causes objects to fall toward Earth could be the same force that keeps the Moon in its orbit.

Assume that Earth's gravitational attraction follows the same inverse-square law. The Moon is approximately 60 Earth radii from Earth's center. Determine the gravitational acceleration of the Moon in terms of the gravitational acceleration g at Earth's surface.

Compare your result with the centripetal acceleration of the Moon, assuming a circular orbit with a period of $T_M = 27.3$ days.

Use $R_E = 6.37 \times 10^6\,\mathrm{m}$ and $g = 9.81\,\mathrm{m/s^2}$.

Solution:

(a) Since the planet completes one full revolution in a period T, it travels a distance equal to the circumference of its circular orbit, $2\pi r$. Therefore, its orbital speed is

$$\boxed{v = \frac{2\pi r}{T}.} \tag{3}$$

(b) The planet undergoes uniform circular motion with centripetal acceleration

$$a_c = \frac{v^2}{r}. \tag{4}$$

By Newton's second law, the magnitude of the force responsible for this acceleration is

$$F = ma_c = \frac{mv^2}{r}. \tag{5}$$

Substituting the expression for the orbital speed gives

$$F = \frac{m}{r}\left(\frac{2\pi r}{T}\right)^2. \tag{6}$$

Simplifying,

$$\boxed{F = \frac{4\pi^2 mr}{T^2}.} \tag{7}$$

(c) For circular orbits, the semi-major axis is equal to the orbital radius, $a = r$. Therefore, Kepler's third law can be written as

$$T^2 = Cr^3, \tag{8}$$

where C is a constant that is the same for all planets orbiting the Sun.

Substituting this expression into the force obtained in part (b), we find

$$F = \frac{4\pi^2 mr}{Cr^3} = \frac{4\pi^2 m}{Cr^2}. \tag{9}$$

Since $4\pi^2/C$ is constant, we conclude that

$$\boxed{F(r) \propto \frac{1}{r^2}} \tag{10}$$

for a planet of fixed mass.

Thus, the force required to maintain planetary orbits decreases with the square of the distance from the Sun.

(d) Let M_E denote Earth's mass. Assuming that Earth's gravitational attraction obeys the inverse-square law, the gravitational acceleration at a distance r from Earth's center has the form

$$g(r) = \frac{K}{r^2}, \tag{11}$$

where K is a constant determined by Earth.

At Earth's surface, $r = R_E$, so

$$g = \frac{K}{R_E^2}. \tag{12}$$

At the Moon's orbital radius, $r_M = 60R_E$, the gravitational acceleration is therefore

$$g_M = \frac{K}{(60R_E)^2} = \frac{1}{60^2}\frac{K}{R_E^2}. \tag{13}$$

Hence,

$$\boxed{g_M = \frac{g}{3600}.} \tag{14}$$

Using $g = 9.81\,\mathrm{m/s^2}$, we obtain

$$\boxed{g_M \approx 2.73 \times 10^{-3}\,\mathrm{m/s^2}.} \tag{15}$$

We can now compare this prediction with the centripetal acceleration required to maintain the Moon's orbit.

The Moon's orbital radius is approximately

$$r_M = 60R_E = 3.822 \times 10^8\,\mathrm{m}, \tag{16}$$

and its orbital period in seconds is

$$T_M = 27.3 \times 24 \times 3600 = 2.35872 \times 10^6\,\mathrm{s}. \tag{17}$$

The centripetal acceleration is

$$a_M = \frac{v_M^2}{r_M} = \frac{4\pi^2 r_M}{T_M^2}. \tag{18}$$

Substituting the numerical values gives

$$\boxed{a_M \approx 2.71 \times 10^{-3}\,\mathrm{m/s^2}.} \tag{19}$$

We therefore find

$$\boxed{a_M \approx g_M.} \tag{20}$$

The agreement between these two independently calculated accelerations supports the hypothesis that the force responsible for the Moon's orbital motion is the same gravitational attraction that causes objects to fall toward Earth's surface.